

Exercise Sheet #6

AC 1

Exercise 1: Sinusoidal Voltage

Given the sinusoidal voltage $v(t) = 50 \cos(30t + 10^\circ)$ V, find:

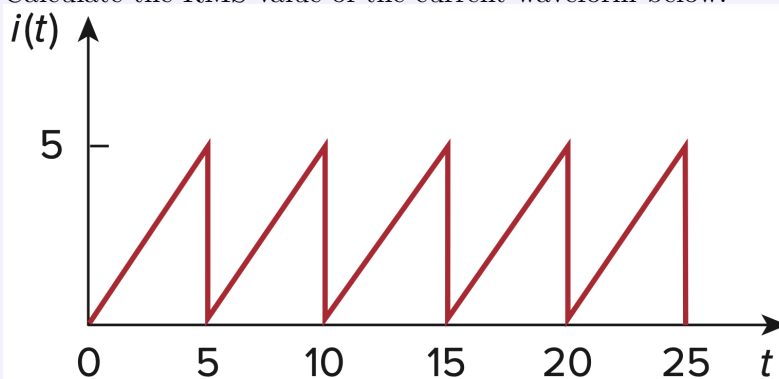
- a) the amplitude V_m
- b) the period T
- c) the frequency f
- d) $v(t)$ at $t = 10$ ms.

Exercise 2: Phase Shift

Given $v_1 = 45 \sin(\omega t + 30^\circ)$ V and $v_2 = 50 \cos(\omega t - 30^\circ)$ V, determine the phase angle between the two sinusoids and which one lags the other.

Exercise 3: RMS Value

Calculate the RMS value of the current waveform below.



Exercise 4: Peak Phasors

For the following signals, find the complex amplitudes corresponding to a cosine reference as well as the rectangular form of the phasors.

- a) $v(t) = 21 \cos(4t - 15^\circ)$ V
- b) $i(t) = -8 \sin(10t + 70^\circ)$ mA

Exercise 5: Phasor Arithmetic

Two voltages v_1 and v_2 appear in series so that their sum is $v = v_1 + v_2$. If $v_1 = 10 \cos(50t - \frac{\pi}{3})$ V and $v_2 = 12 \cos(50t + 30^\circ)$ V, find v .

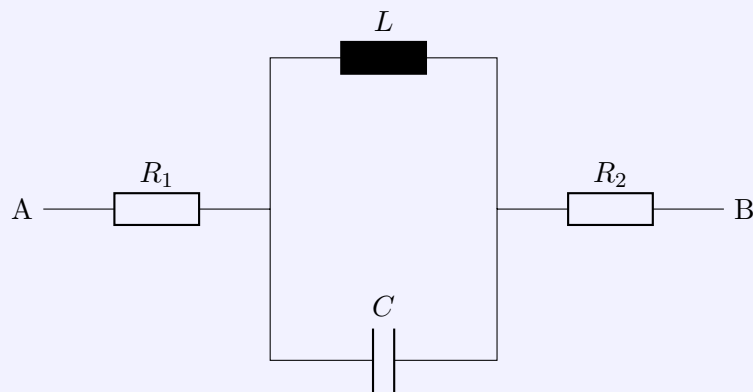
Exercise 6: AC Circuit

A series RLC circuit has $R = 80\ \Omega$, $L = 240\text{ mH}$, and $C = 5\text{ mF}$.

If the input voltage is $v(t) = 10 \cos(2t)\text{ V}$, find the current flowing through the circuit.

Exercise 7: Equivalent Impedance

Given the AC network with $R_1 = 10\ \Omega$, $R_2 = 20\ \Omega$, $L = 50\text{ mH}$, $C = 100\ \mu\text{F}$, $\omega = 1000\text{ rad/s}$:

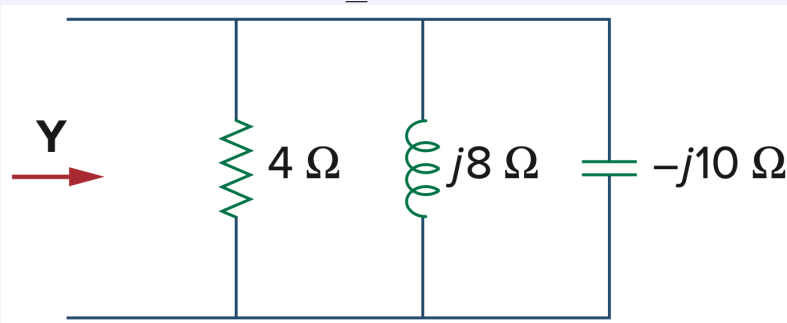


Determine:

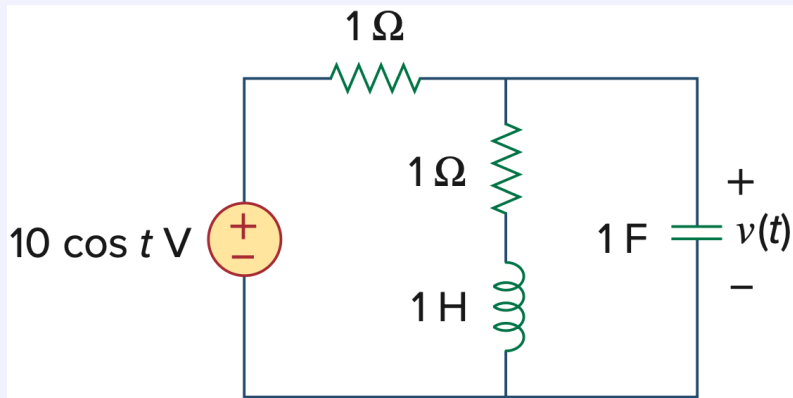
1. the complex impedances of the individual components,
2. the complex equivalent impedance between terminals A and B,
3. the magnitude and phase of the equivalent impedance.

Exercise 8: Equivalent Admittance

Determine the admittance $\underline{Y}$ for the circuit shown below.

**Exercise 9: RLC Circuit**

Find $v(t)$ in the circuit below.



Exercise 10: Phase-shifted Source Current

Determine the value of $i_s(t)$ in the circuit below.

